

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/D_DEB_SAAWKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3303, C2 = 3666, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.0001,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1, 0.001,    2);
Distrib (kap,         Uniform,                0,        1);
Distrib (kapA,        Uniform,                0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,   0.3);
Distrib (K,           Uniform,                0.00001,   2);
Distrib (dv,          TruncNormal_cv,        2,      0.5, 0,      10);
Distrib (wE,          LogUniform,          0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

#f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

Fm.1. pAm.1. r_pAm_pM.1. v.1. kap.1. kapA.1.

Result_mode	0.002944000	174.72500	0.8609600	0.19262700	0.93422100	0.56604400
2.5%	0.000929474	152.02550	0.6809205	0.15532900	0.88459300	0.43589100
97.5%	0.005218970	192.40335	0.9137580	0.22636800	0.94980500	0.60201700
Min	0.000743968	104.83800	0.5205580	0.12708900	0.12674200	0.21707400
Max	0.320981000	216.44600	1.0000400	0.26149400	0.95599300	0.69570300
Result_mean	0.003556793	171.10609	0.7906414	0.19001211	0.92022034	0.51847024
Result_sd	0.003012470	10.52978	0.0573568	0.01798793	0.01739831	0.04117091
	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	K.1.	dv.1.	
Result_mode	90.72450	27.141400	0.009102830	0.02257400	4.1655300	
2.5%	75.18800	23.841130	0.006053482	0.00826841	3.2020100	
97.5%	136.16600	37.603300	0.011636300	0.04220520	5.4746100	
Min	60.57770	20.559000	0.000646208	0.00607311	0.5220400	
Max	176.36800	45.472500	0.289479000	1.59821000	6.0374800	
Result_mean	102.70932	29.962885	0.008803699	0.02790131	4.2185172	
Result_sd	14.90937	3.439699	0.002189514	0.01435645	0.5639362	
	wE.1.	Sigma_W.1.				
Result_mode	2.359430e-09	0.12668300				
2.5%	1.091170e-09	0.11874420				
97.5%	5.426420e-06	0.14989300				
Min	1.003290e-09	0.10461900				
Max	9.851330e-06	5.52230000				
Result_mean	6.036755e-07	0.13345170				
Result_sd	1.355887e-06	0.05250067				

Number of observations, n = 277
 Number of parameters, k = 12
 Log-likelihood, LL = -33.0155
 BIC = 133.5192

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.99	3.41
E_coc.1.	1.83	3.00
Ehp.1.	1.03	1.11
Fm.1.	4.06	8.14
K.1.	4.10	8.37
kap.1.	1.46	2.21
kapA.1.	1.26	1.76
pAm.1.	1.55	2.42
r_pAm_pM.1.	1.11	1.30

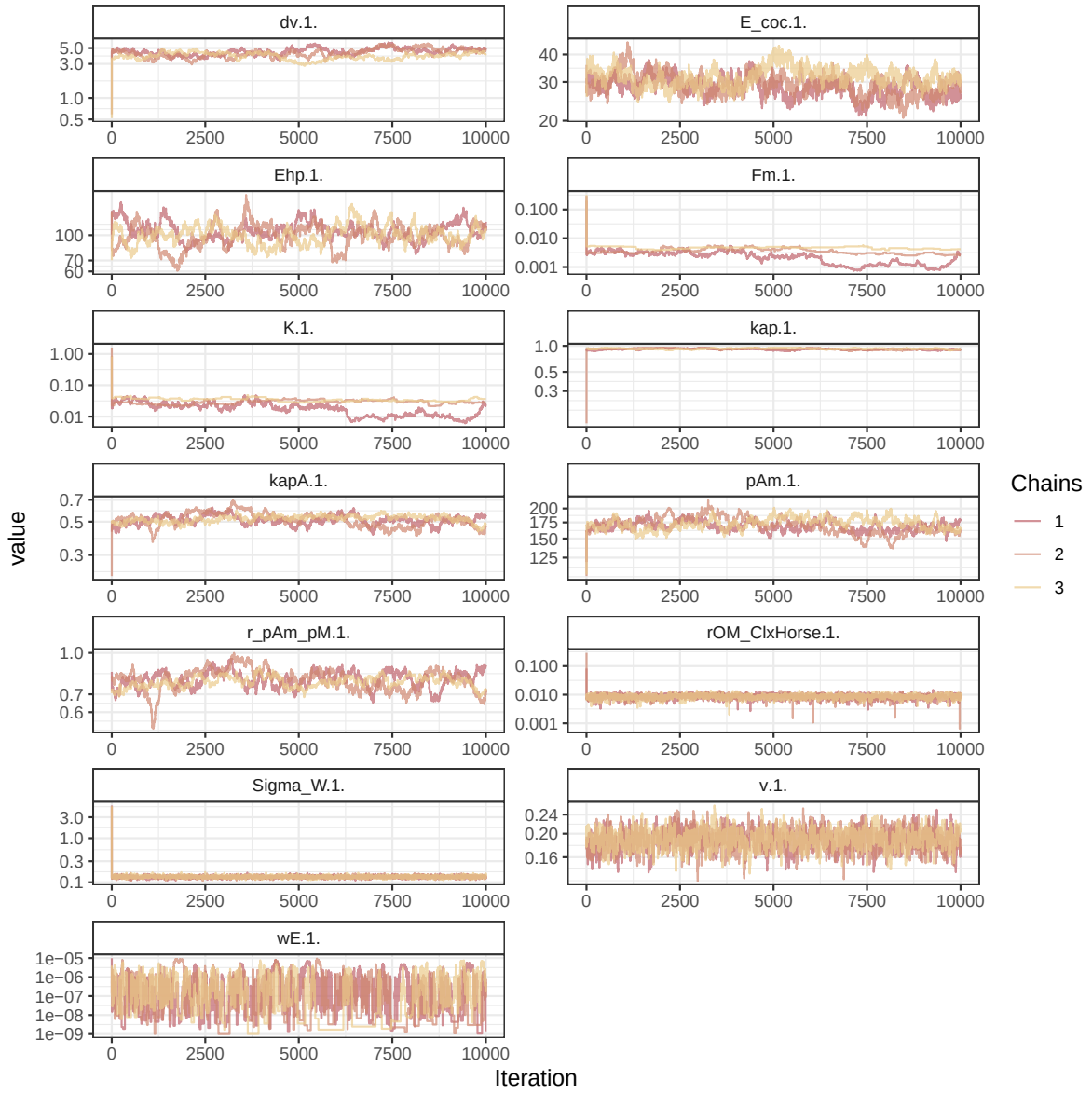


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

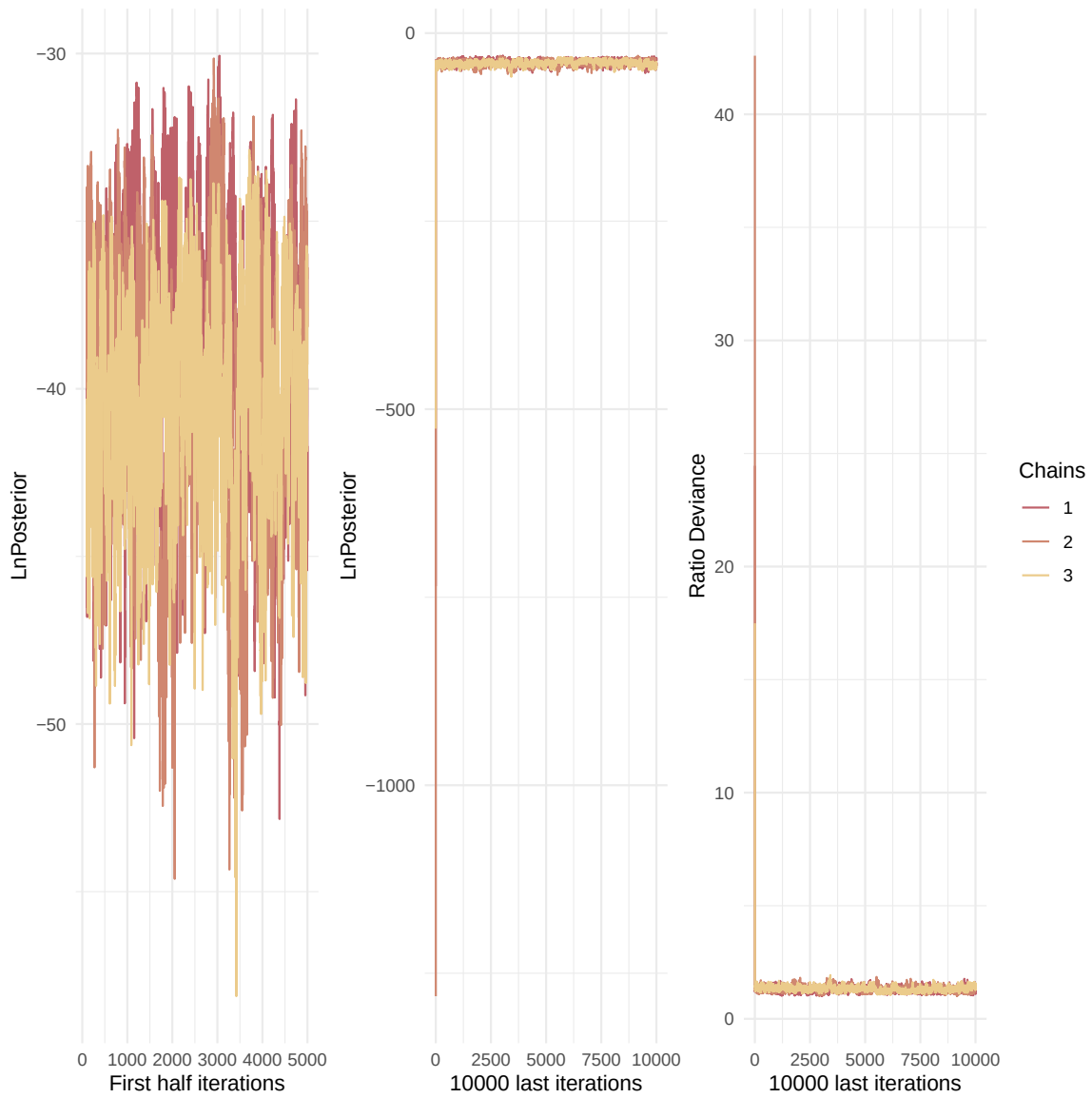


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

rOM_ClxHorse.1.	1.02	1.07
Sigma_W.1.	1.00	1.02
v.1.	1.01	1.03
wE.1.	1.02	1.06

Multivariate psrf

3.38

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

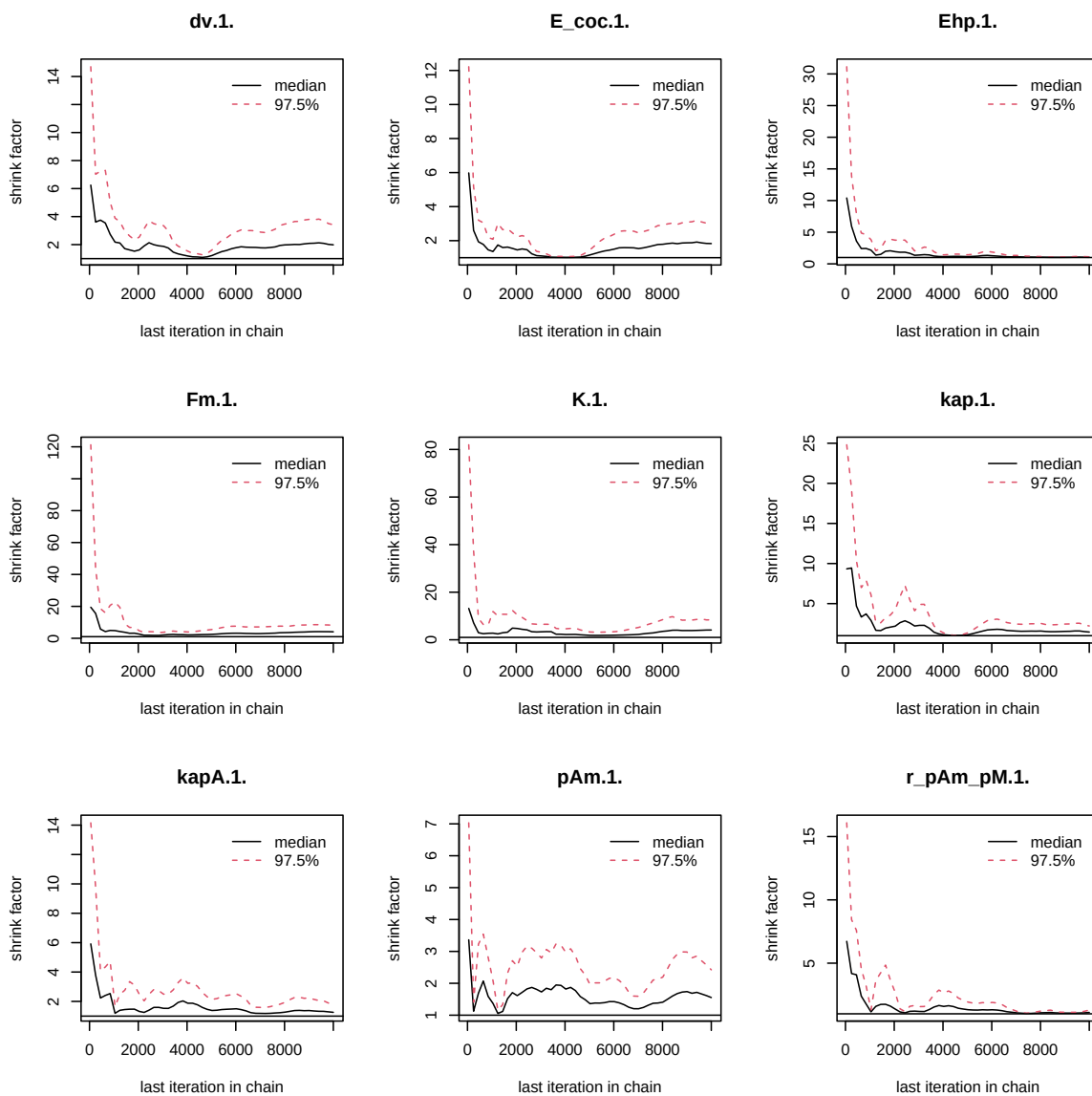


Figure 3: Chains convergence.

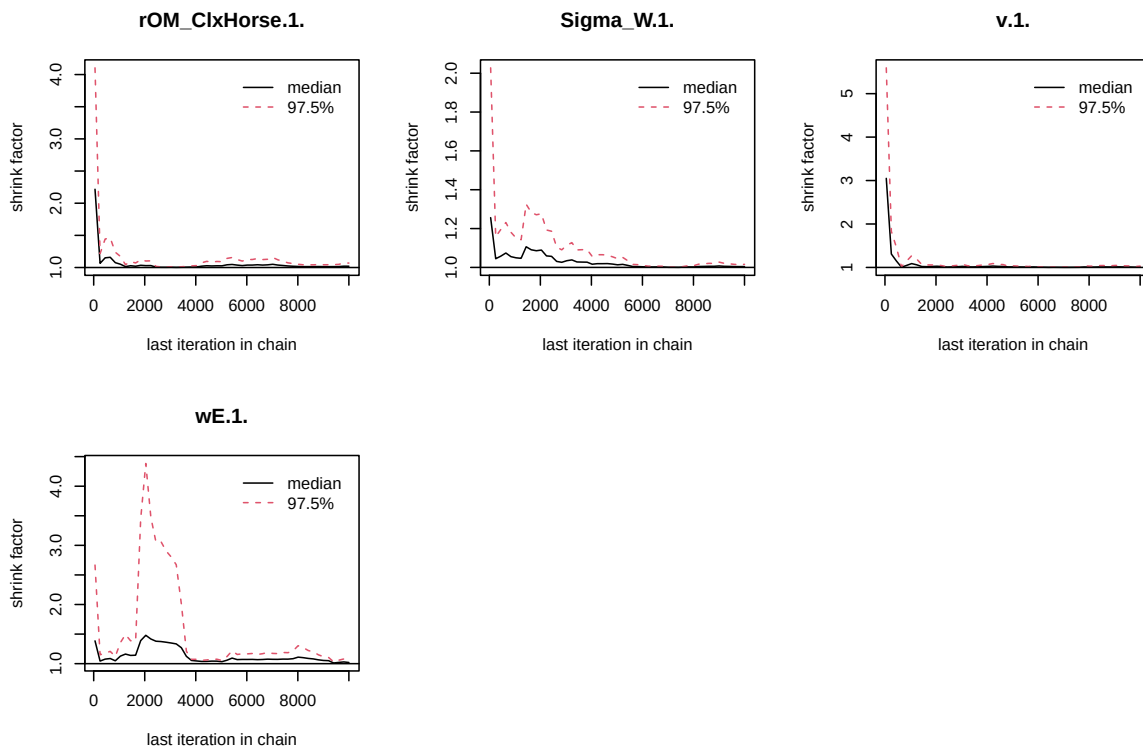


Figure 4: Chains convergence.

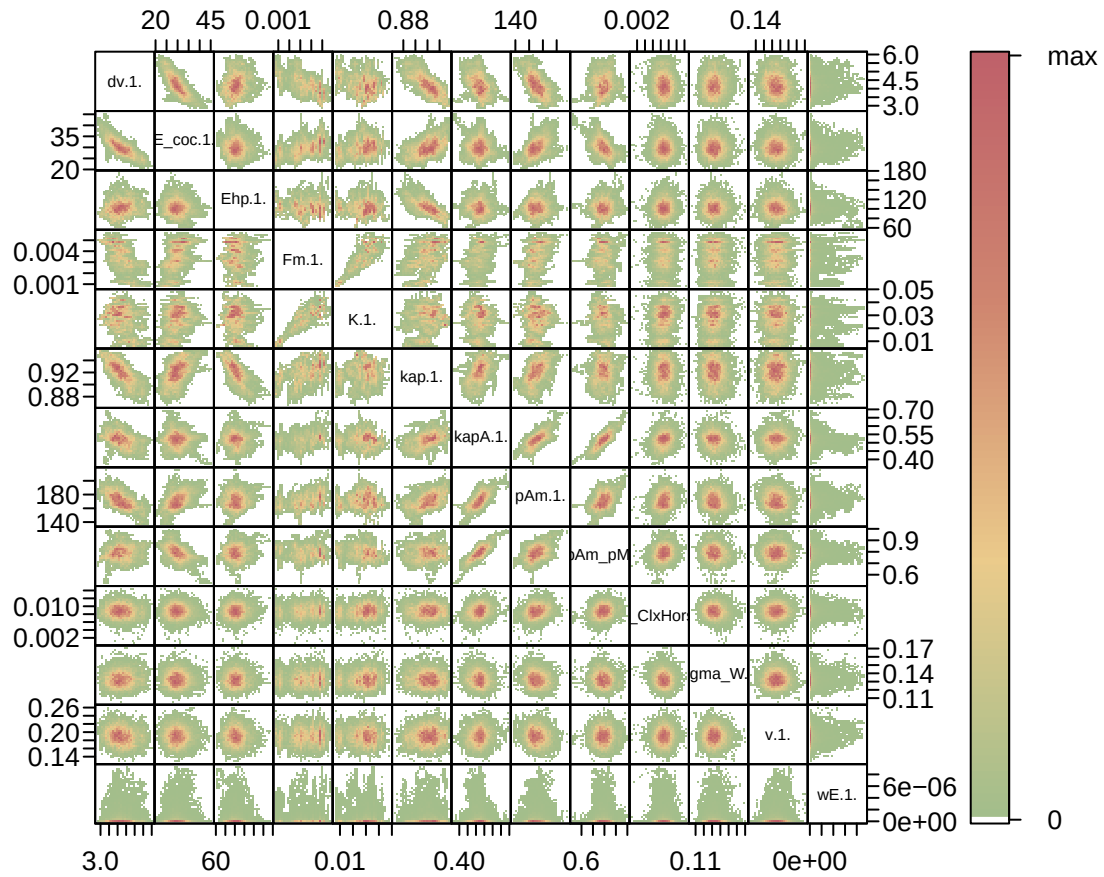


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

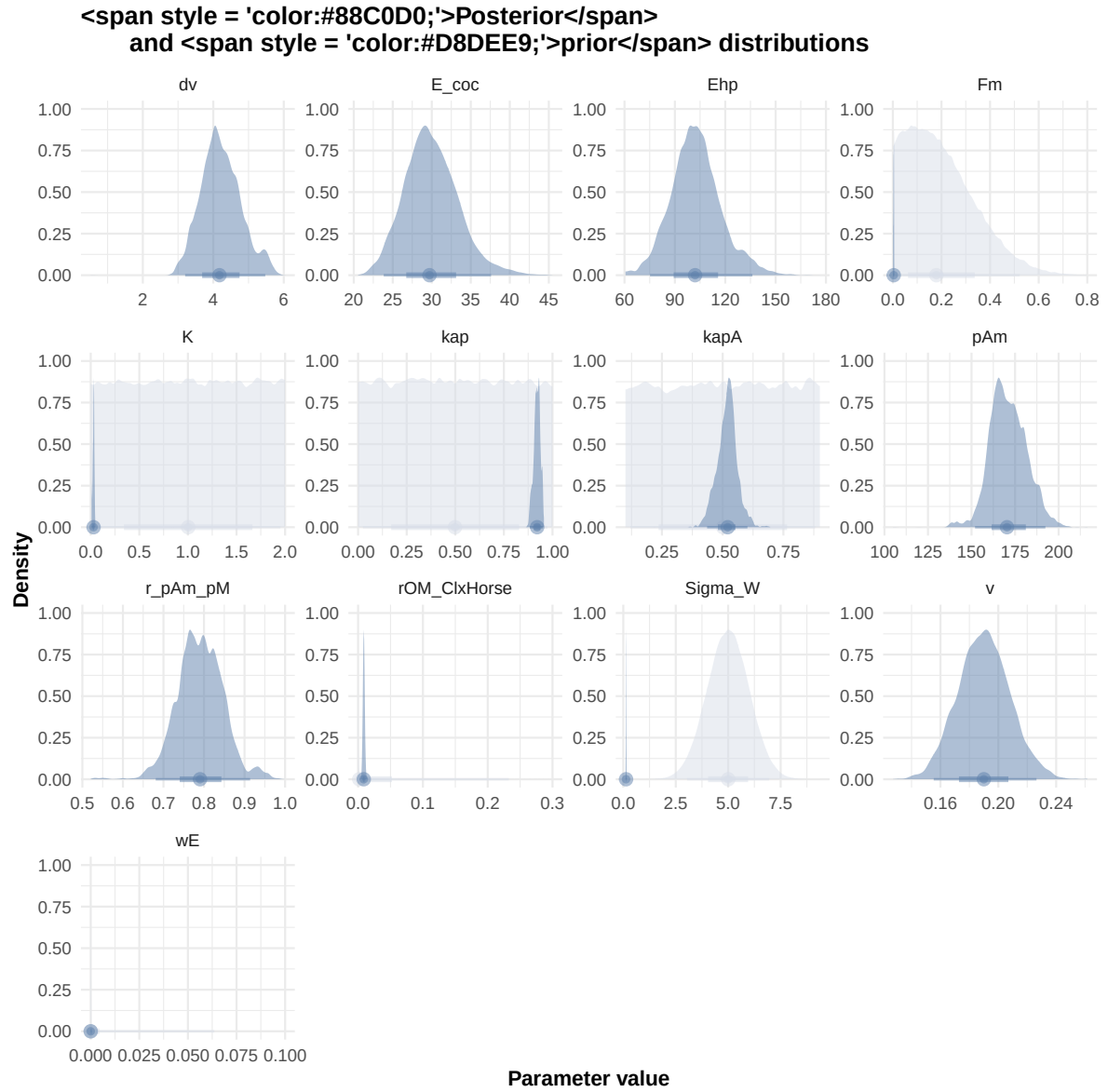


Figure 6: Distributions of the priors and the posteriors.

```

"Bart2020",
#"Gollot2026_lufa"
"Gollot2026_dens_D1",
"Gollot2026_dens_D2",
"Gollot2026_dens_D2b",
"Gollot2026_dens_D5",
"Gollot2026_dens_D7"
)

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\.\\.$|\\.1\\.\\.", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,
  kapreal,
  freal,
  Qf_soil,
  Qf_horse,
  X"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcali DEB_sim_full.in

```

```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

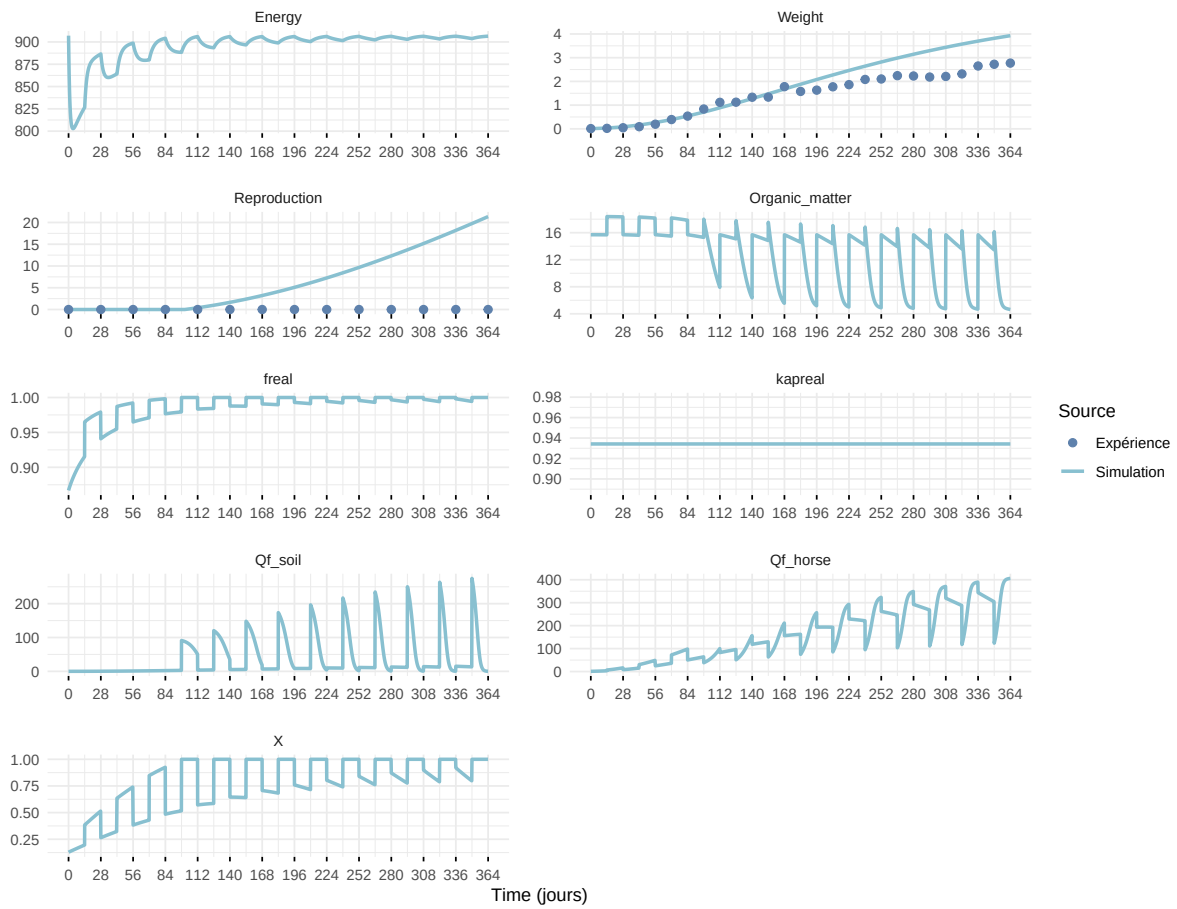
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected join
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =

"many-to-many" to silence this warning.

Expérience : Bart2019_XP1_1



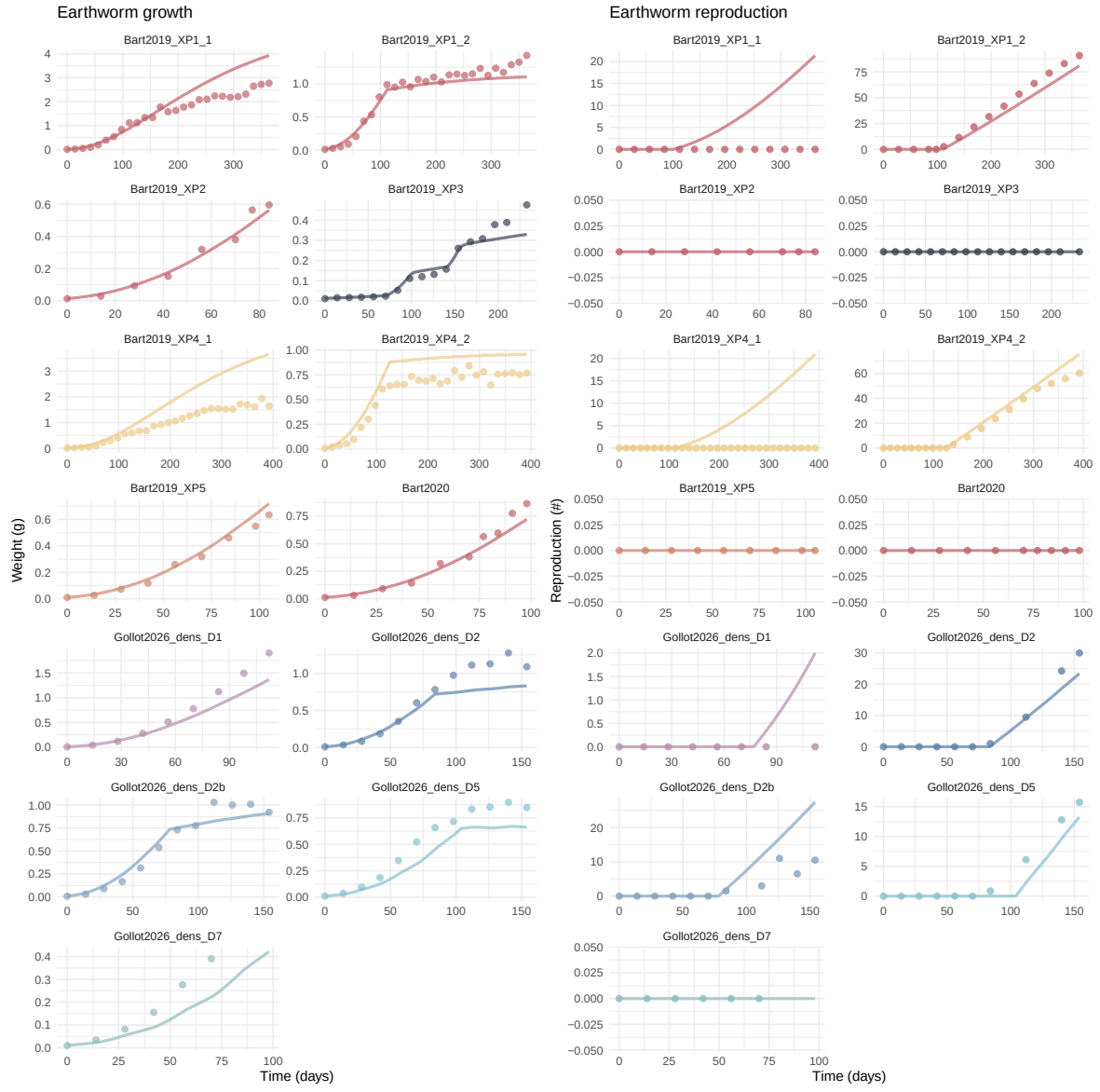
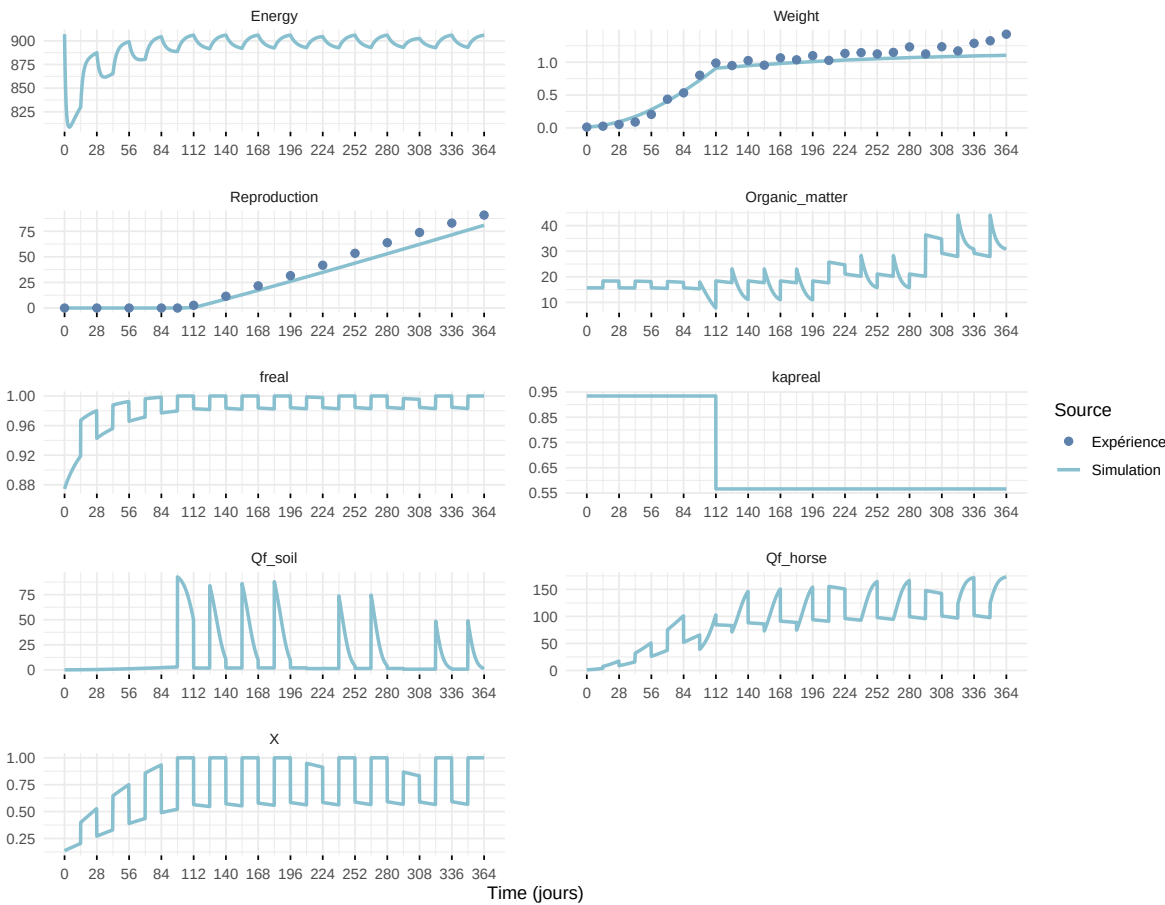
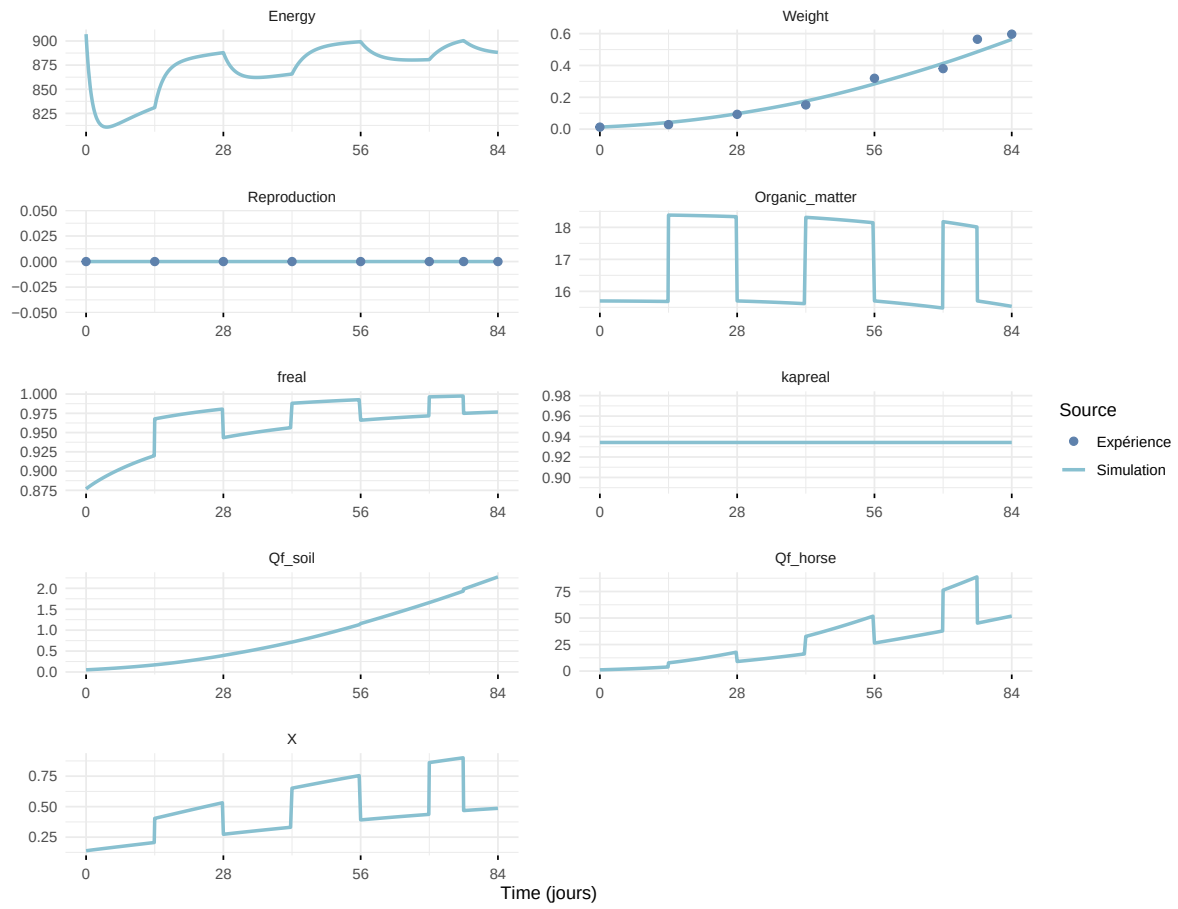


Figure 7: Simulations Weight and Reproduction.

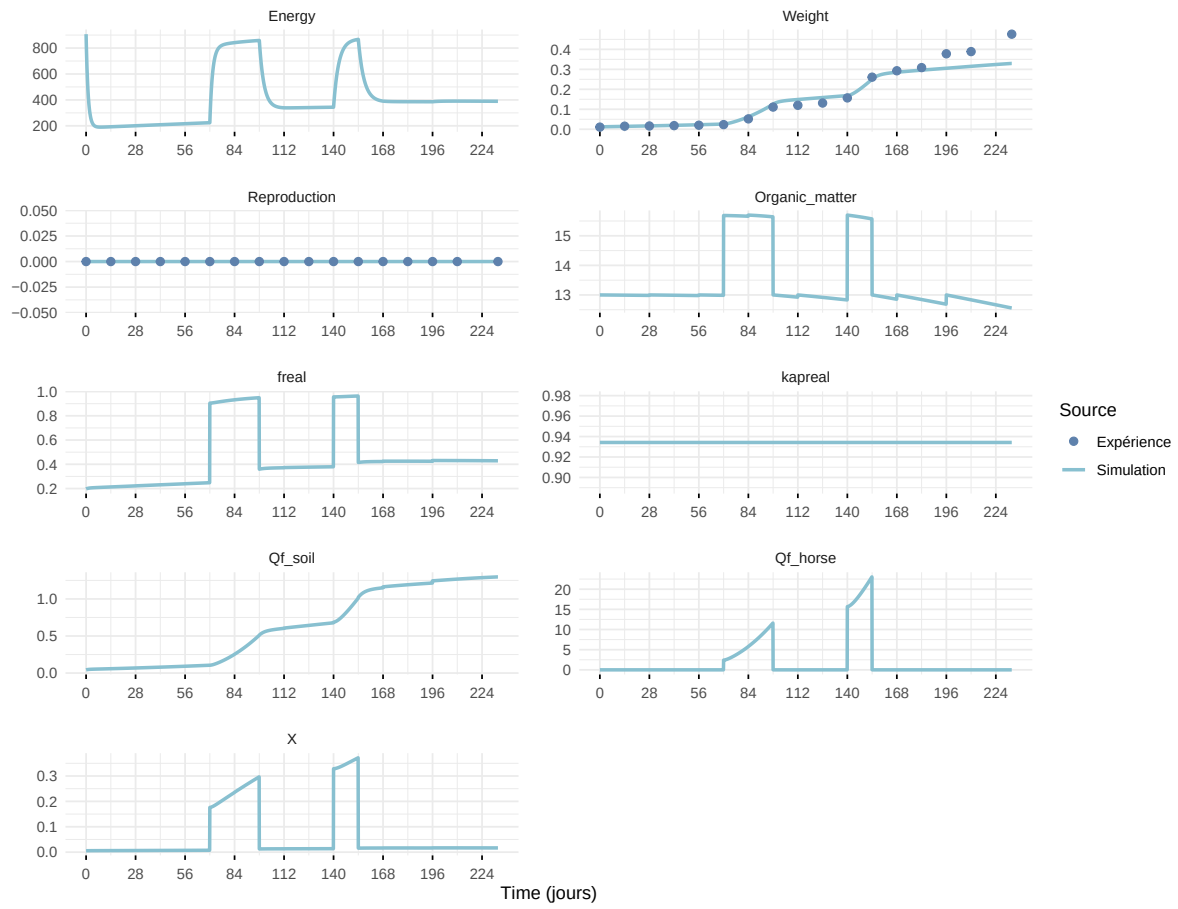
Expérience : Bart2019_XP1_2



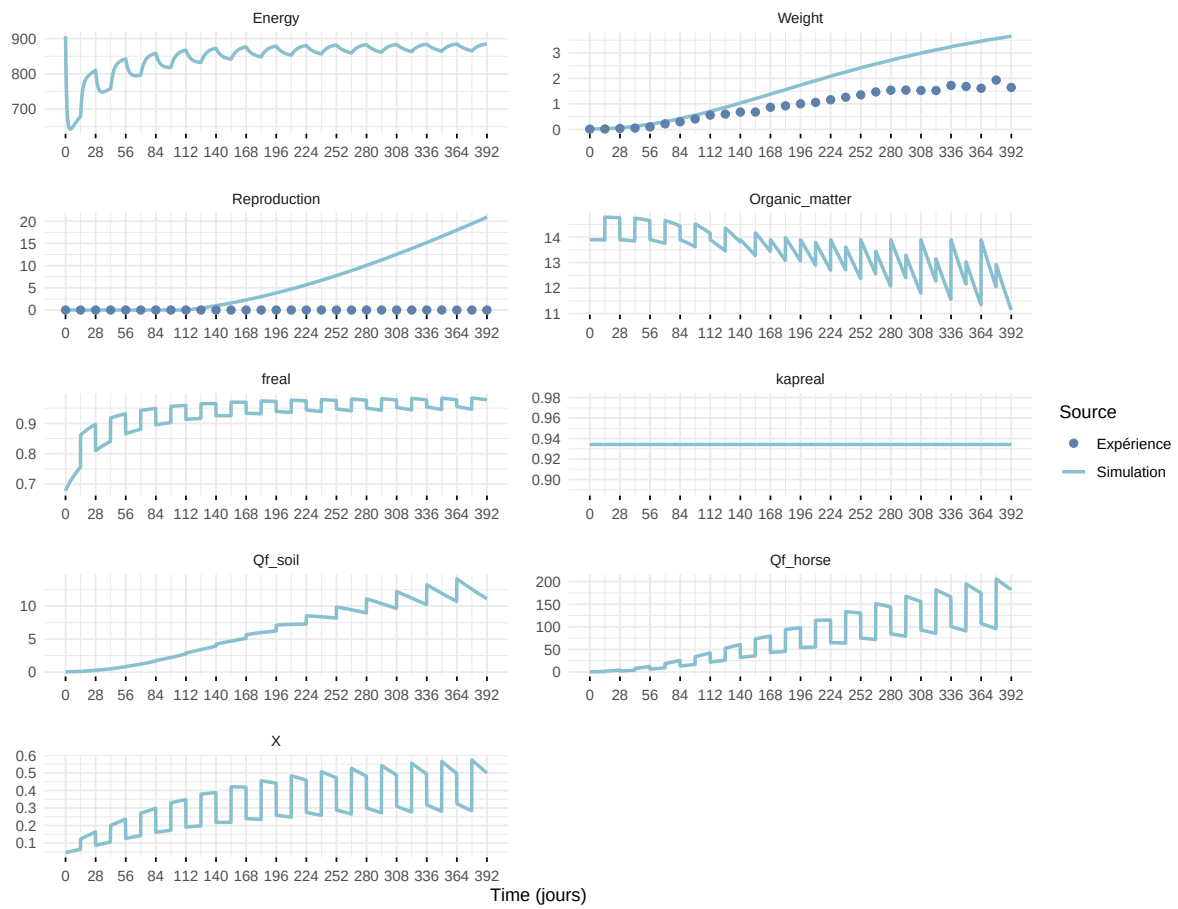
Expérience : Bart2019_XP2



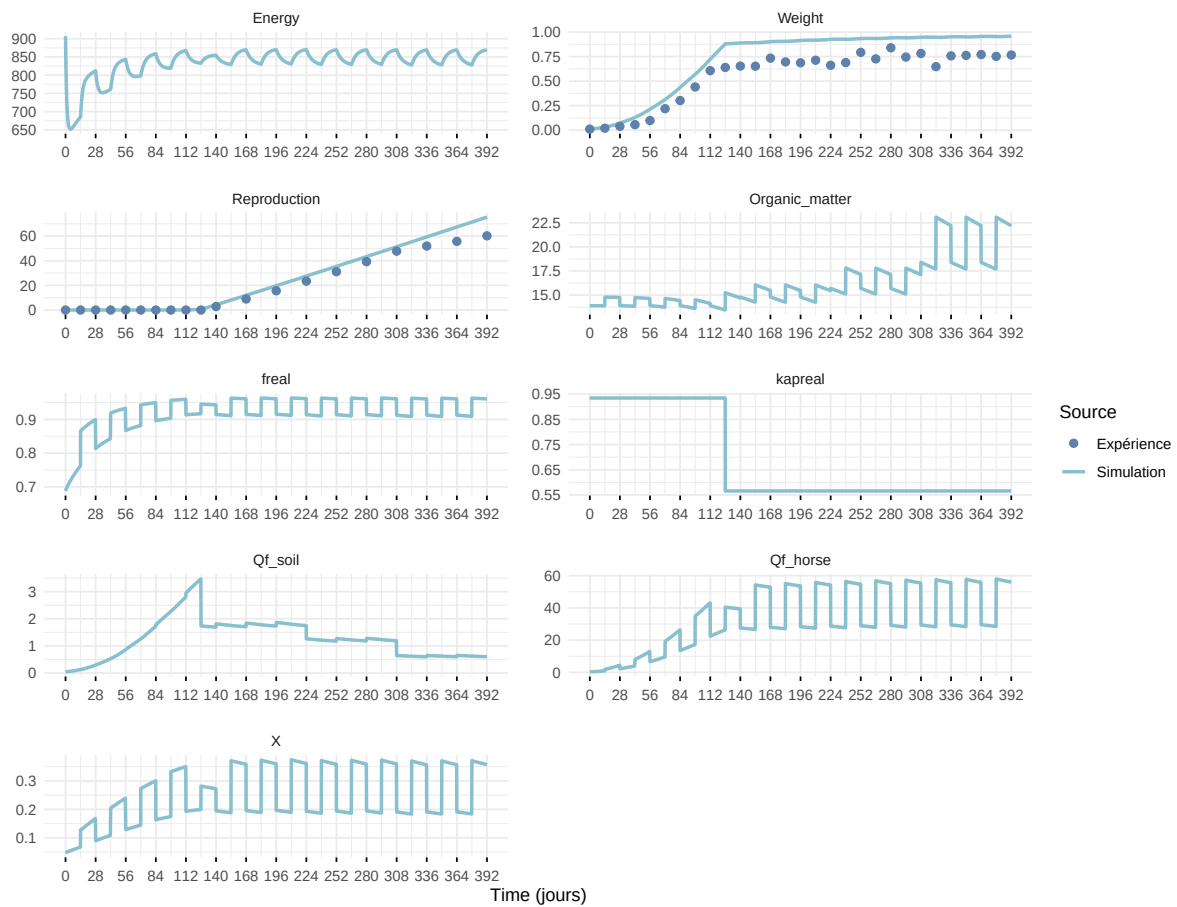
Expérience : Bart2019_XP3



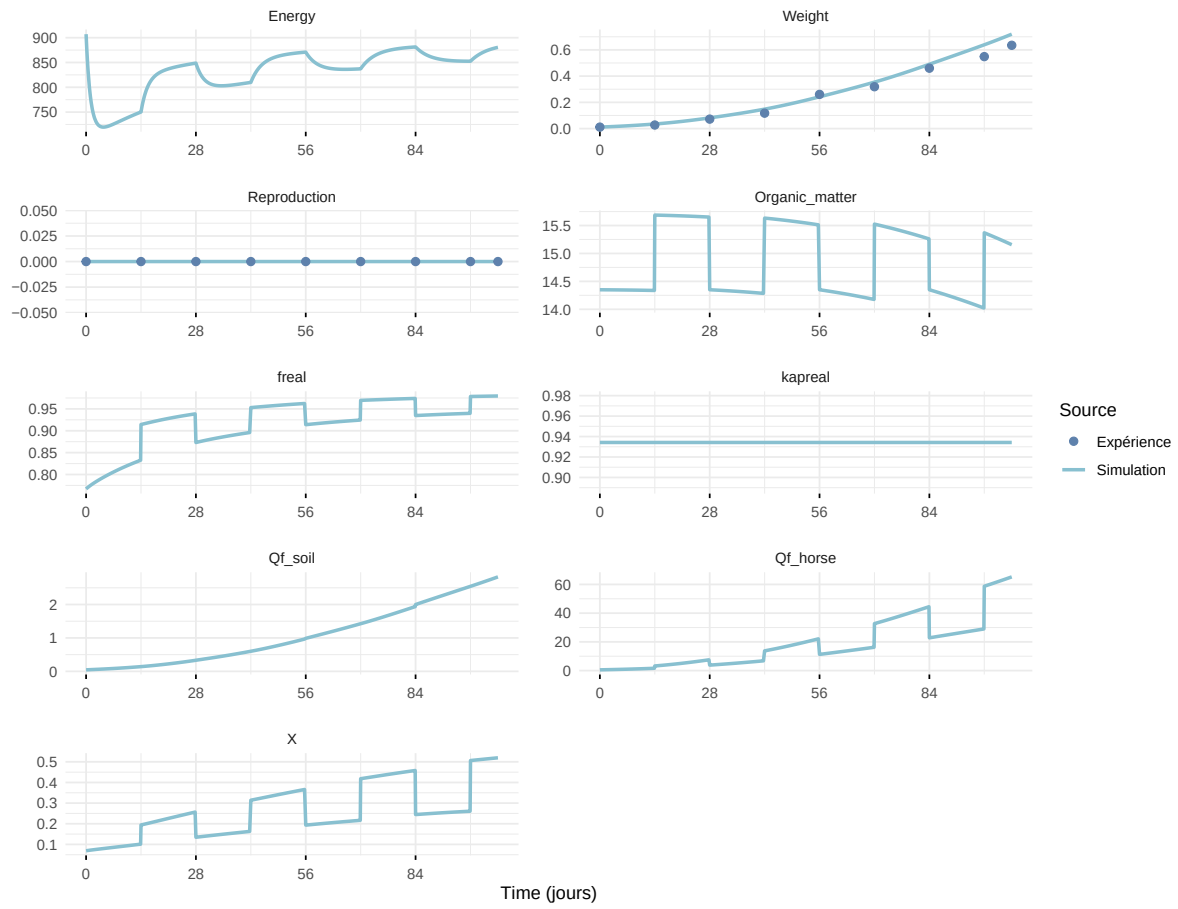
Expérience : Bart2019_XP4_1



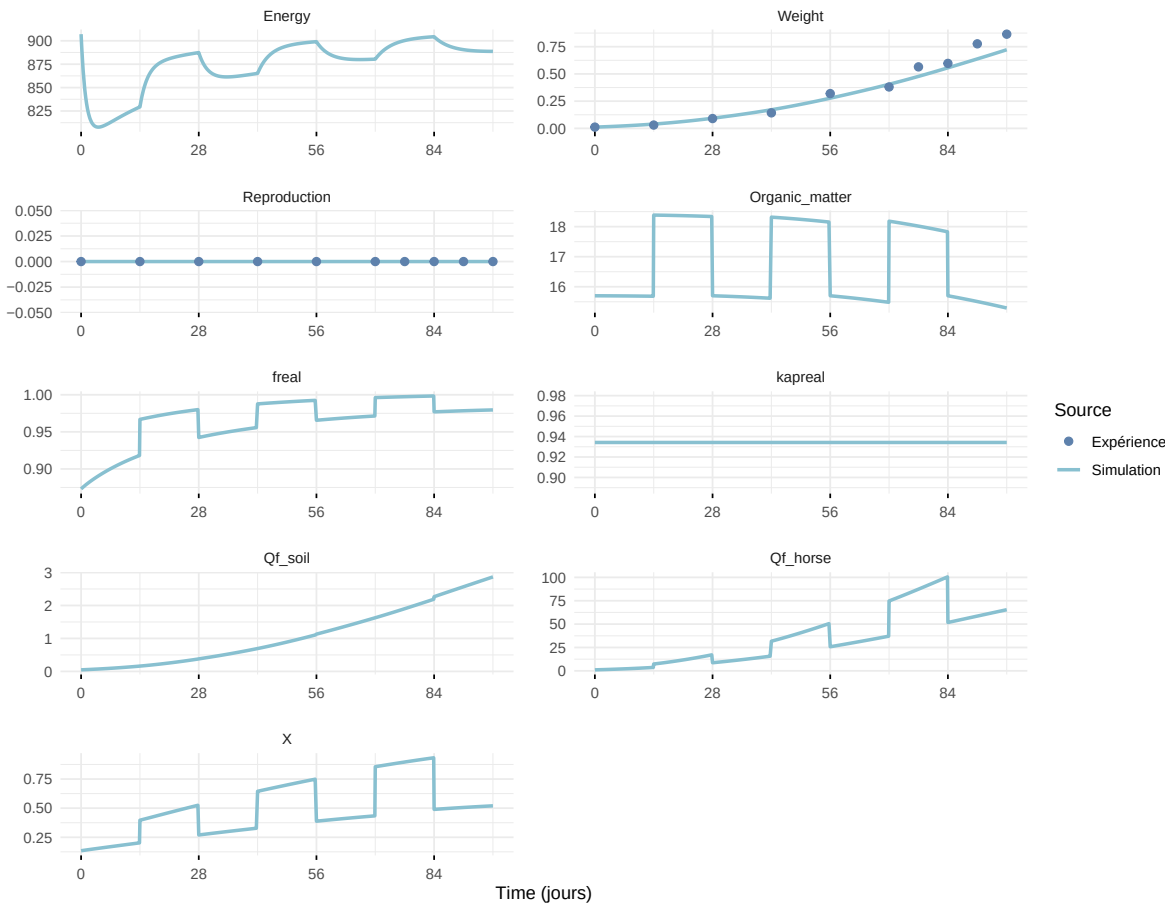
Expérience : Bart2019_XP4_2



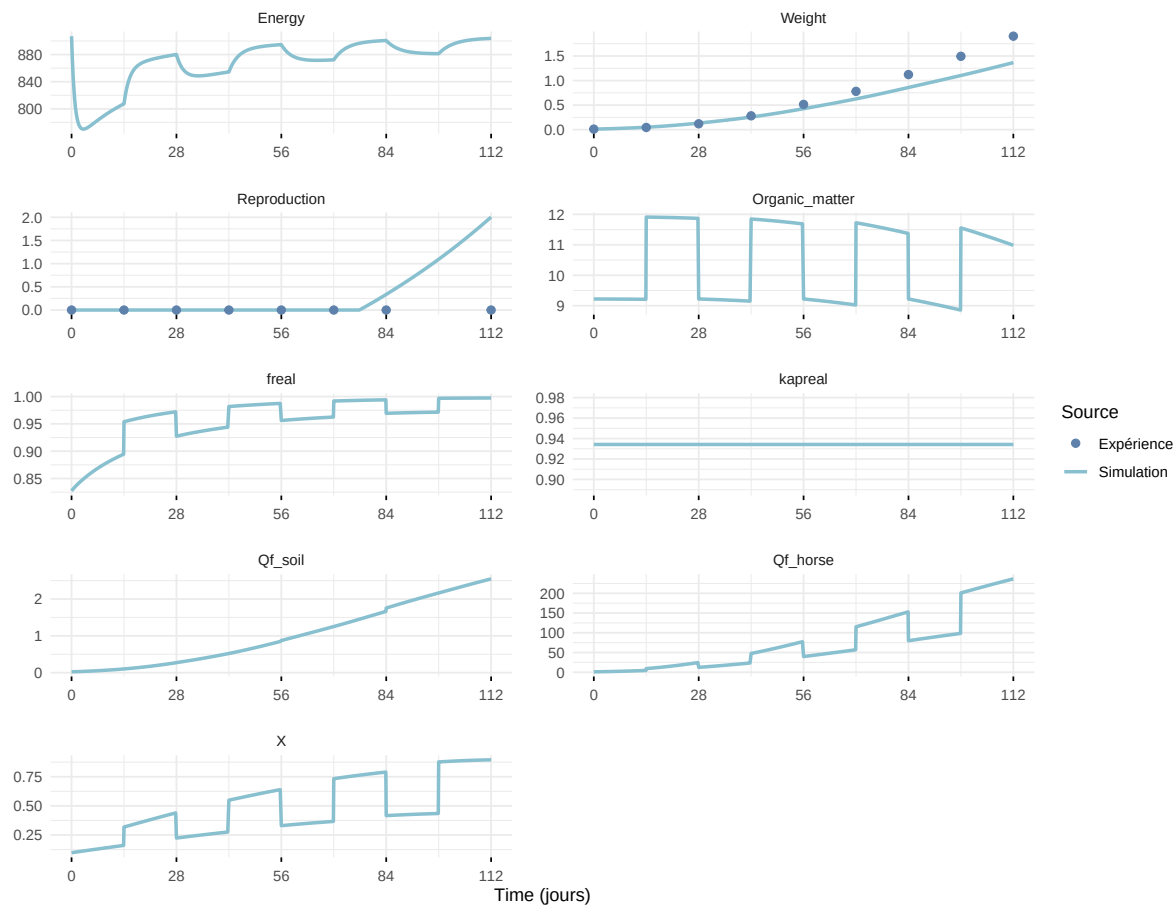
Expérience : Bart2019_XP5



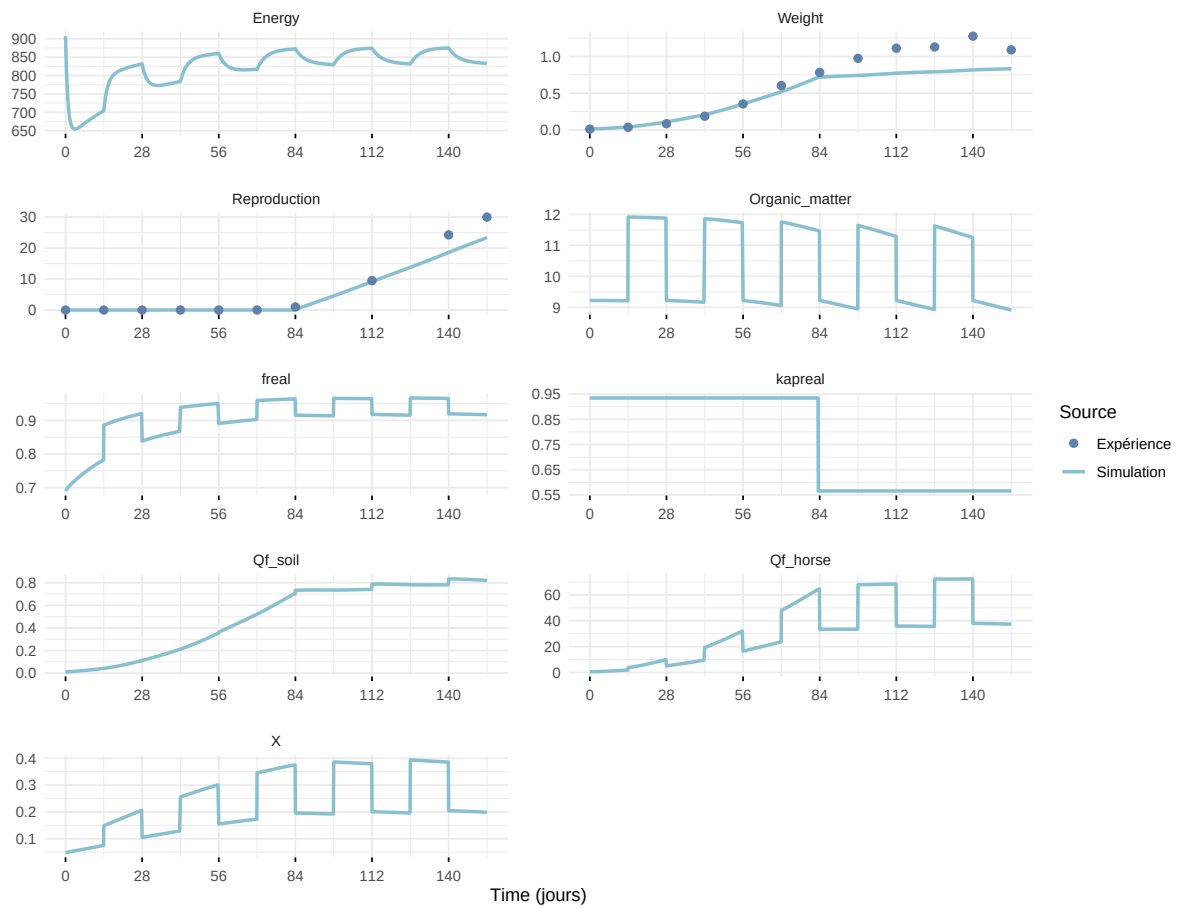
Expérience : Bart2020



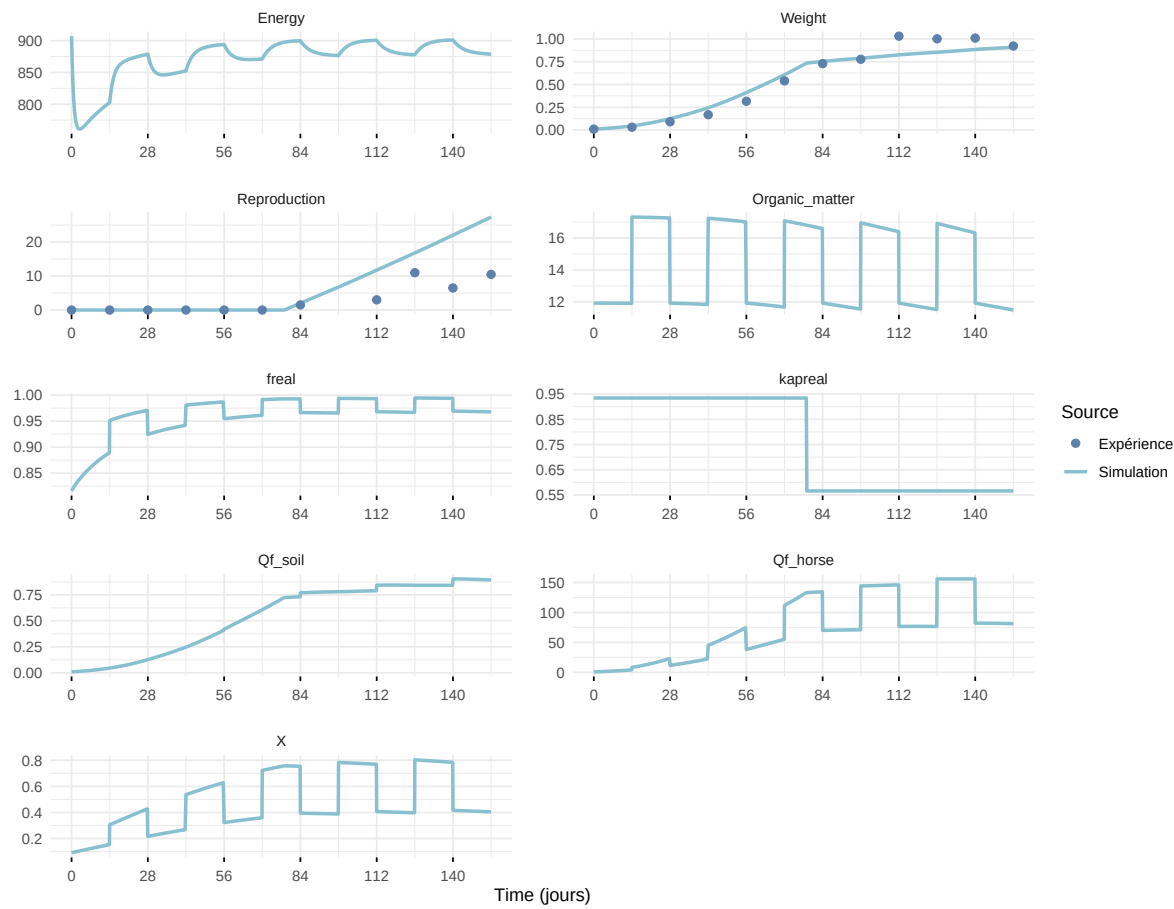
Expérience : Gollot2026_dens_D1



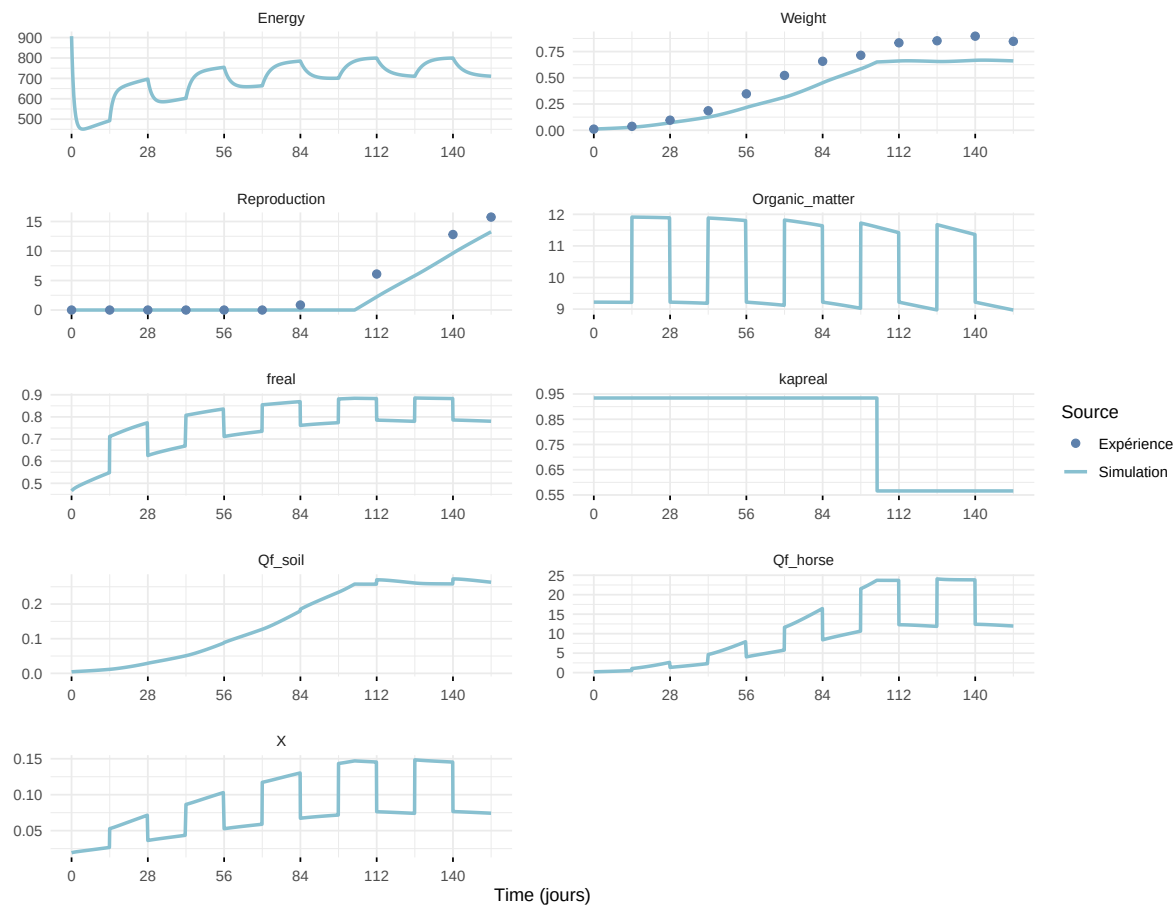
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

